

Credit Card Fraud Detection using ML and Generative AI

This report includes the complete source code provided by the user followed by project information.

Complete Python Code

```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    roc_auc_score,
    confusion_matrix,
    classification_report,
    ConfusionMatrixDisplay
)

# Load dataset
df = pd.read_csv("creditcard.csv")

print("Dataset Shape:", df.shape)
print(df.head())

# Check missing values
print("\nMissing Values:")
print(df.isnull().sum())

# Class distribution
print("\nFraud vs Normal:")
print(df["Class"].value_counts())

df["Class"].value_counts().plot(kind="bar")
plt.title("Fraud vs Normal Transactions")
plt.xlabel("Class")
plt.ylabel("Count")
plt.show()

X = df.drop("Class", axis=1)
y = df["Class"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
y_prob = model.predict_proba(X_test)[:, 1]

print("\nAccuracy :", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred))
print("Recall   :", recall_score(y_test, y_pred))
print("F1 Score :", f1_score(y_test, y_pred))
print("ROC AUC  :", roc_auc_score(y_test, y_prob))

print("\nClassification Report")
print(classification_report(y_test, y_pred))

cm = confusion_matrix(y_test, y_pred)
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=["Normal", "Fraud"])
disp.plot(cmap="Blues")
plt.title("Confusion Matrix")
plt.show()

importance = pd.Series(model.feature_importances_, index=X.columns)
importance.sort_values(ascending=False).head(10).plot(kind="barh", figsize=(8,5))
```

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plt.title("Top 10 Important Features")  
plt.show()
```

Generative AI can generate synthetic fraud samples, explain model predictions, create investigation reports, summarize suspicious transactions, and assist fraud analysts through conversational AI. Technologies: Python, Pandas, Matplotlib, Scikit-learn, Random Forest, Git/GitHub, Jupyter Notebook, Generative AI (LLMs).